Bachelor of Computer Science

Subject Code: CIS3201
Subject Name: COMPUTER COMMUNICATION AND NETWORKS
Assessment – 20 %

Group Assignment
(Project)

Group Member

No	Student ID	Full Name	Leader/Member
1	I25036157	Amritha Lakshmidaran	Leader
2	125037156	Deepak Senthilkumar	Member
3	125036891	Adharsh Manikandan kumari Bindhu	Member
4	125037033	Anish Kanna Jegadheesan	Member
5	125035985	Ramshah Sualeh	Member

MARKING PAGE

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CIS3201 COMPUTER COMMUNICATION AND NETWORKS - GROUP ASSIGNMENT RUBRIC

Criteria	Excellent	Good	Satisfactory	Needs Improvement	Poor	Max Mark	Score
Network Design	21-25	16-20	11-15	6-10	1-5		
	Comprehensive, innovative, and well-documented design with clear diagrams and rationale.	Detailed and logical design with clear diagrams and rationale.	Basic design with some diagrams and explanations.	Incomplete design with unclear diagrams and explanations.	Poor or missing design, diagrams, and explanations.	25	
Device Configuration	21-25	16-20	11-15	6-10	1-5		
	Thorough and accurate configuration of all devices, with detailed documentation.	Accurate configuration of most devices, with clear documentation.	Basic configuration of devices, with some documentation.	Incomplete configuration of devices, with limited documentation.	Poor or missing configuration and documentation.	25	
Internet Connectivity	5	4	3	2	1		
	Comprehensive plan for internet connectivity with clear, detailed diagrams and rationale.	Clear plan for internet connectivity with good diagrams and rationale.	Basic plan for internet connectivity with some diagrams and rationale.	Incomplete plan for internet connectivity with unclear diagrams and rationale.	Poor or missing plan for internet connectivity, diagrams, and rationale.	5	
Services and Applications	13-15	10-12	7-9	4-6	1-3		
	Thorough and well-integrated plan for services and applications, with detailed documentation.	Clear plan for services and applications, with good documentation.	Basic plan for services and applications, with some documentation.	Incomplete plan for services and applications, with limited documentation.	Poor or missing plan for services and applications, and documentation.	15	
Testing, Validation & Troubleshooting	13-15	10-12	7-9	4-6	1-3		
	Comprehensive and detailed testing, validation, and troubleshooting plan.	Clear and detailed testing, validation, and troubleshooting plan.	Basic testing, validation, and troubleshooting plan.	Incomplete testing, validation, and troubleshooting plan.	Poor or missing testing, validation, and troubleshooting plan.	15	
Presentation	13-15	10-12	7-9	4-6	1-3		
	Exceptionally clear and well-presented report; free of errors.	Clear and well-presented report; minor errors.	Satisfactory presentation; some errors.	Presentation needs improvement; multiple errors.	Poorly presented report; numerous errors.	15	

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NETWORK DESIGN

The network is expected to provide us with a rapid, safe, and scalable installation, enabling the entire organization to engage in conversation with one another, move things around, and utilize resources at will. My hierarchical, modular plan was chosen to be more troubleshooting-friendly, so traffic can keep running, and we can add more stuff later. I have discussed in detail topology, segmentation, device selection, connectivity medium, redundancy, security, and monitoring (see refs).

1. Network Architecture Overview.

The network employs a three-tier hierarchical design with three layers, namely core, distribution, and access, which divides traffic management, isolates faults, and scales well.

a. Core Layer

The backbone is the high-speed one, which is the core:

- Manages inter-departmental and high-volume traffic.
- VPN Routing and load balancing by means of high-throughput Layer-3 switches.
- Make sure that it is effectively used to block single points of failure by means of redundant links.
- Dynamically routes to achieve good route decisions.

b. Distribution Layer

Connection policy. This distribution layer provides policy-based access between the access and core layers:

- Aggregates multiple access switch traffic.
- Adapts VLAN routing, access control, and traffic prioritization.
- Takes sensitive and non-sensitive segregation to enhance safety.
- Enables the link aggregation to enhance the bandwidth and redundancy.

c. Access Layer

The access layer is connected with the end-user devices, and it interconnects computers, IP phones, printers, and wireless access points.

- Divides users using VLANs based on department or functions.
- port-security Enhances device security by ensuring port access.
- Intensive Power over Ethernet (POE) to IP phones and IP access points.

2. VLAN Strategy and Network Segmentation.

Segmentation allows maintaining the network as secure, fast, and manageable. It is designed with VLANs on a per-department basis (HR, Finance, IT, etc).

- Server VLAN: Vital servers are separated from user devices.
- Voice VLAN: VoIP exclusively to support the quality of calls.
- Guest VLAN: Allows people who have visited to use the internet without interfering with internal resources.
- VLAN Management: To add security to a network for management.

Benefits of Segmentation:

Limits broadcast domains- trims out pointless traffic.

- Locks out production of sensitive traffic.

Next-generation firewalls provide support for VPN, deep packet inspection, and intrusion detection. It prevents threats within the organization and outside.

d. Wireless Access Points

- APs of enterprise-grade and WPA3.
- Facilitate smooth roaming, load balancing, and isolation of guests' networks.
- centrally controlled to manage and monitor.

e. Servers

We installed file, authentication, application, and database servers. Using one central server makes everything easier to maintain, secure, and back up.

DEVICE CONFIGURATION

According to our network diagram, it is the device configuration that keeps making all our networking gear work in the way it is supposed to. It maintains a systemic consistency, security, and reliability throughout the entire campus. This section is a division of the primary routers, switches, firewall, server, and wireless equipment workload.

1. Router Configuration

Internal routing is done by routers that are connected to the outside world:

a. Basic System Setup

The router has a clear hostname, which can be spotted in the logs; it is recommended to provide it.

- Activate secure access to administration based on encrypted data (SSH, HTTPS).
- configure user authentication and RBAC such that only appropriate people are able to do appropriate things.

b. Internal Routing

- It allows routing between internal VLANs through dynamically created protocols such as OSPF or EIGRP.
- Implement routing policies in order to maintain a fluid traffic flow.
- Adding the distribution and core layers to the static or default routes

c. Internet Connectivity

- Interconnect external interfaces with the IPs of the ISP.
- Configure outbound ACLs to prevent unauthorized traffic.
- Use NAT in order to allow external users to access the internet safely.

d. Security Controls

- It only allows connections to a read-only network.
- It disables services and interfaces that are not needed and reduces the attack surface.
- Packet filtering and firewall filters should be used to ingate ominous or this traffic that is not essential.

2. Switch Configuration

The Switching maintains the stability level of the network at the access, distribution, and core levels:

a. VLAN Implementation

- Develop a VLAN for one department or functional unit.
- Pin ports to the right VLANs.
- Trunk connections are done with set trunks to connect across switches carrying several VLANs.

b. Port Security

- Have a maximum number of devices on each port.
- Issue a security measure when an unauthorized device is detected, and Unused ports need to be shut down to minimize entry.

c. Inter-VLAN Communication

- Core/distribution switch routing- Layer-3 routing is enabled.
- Use ACL to put traffic in between departments under control.
- Be effective in routing and implementing policy.

d. Redundancy

This concept represents the relationship of ideas whereby one idea is said to be governed by another.

Redundancy and Link Management:

This concept posits the relationship of ideas whereby an idea is held to be subject to another.

- Bandwidth and failover: Use link aggregation.
- install STP/RSTP in order to keep loops out of the network.

3. Firewall Configuration

The alarms that block both incoming and outgoing traffic are firewalls:

a. Policy Rules

Decision on what traffic to allow or deny.
Only open basic services (web, mail, VPN).

- Block any suspicious traffic tracks.

b. Intrusion Prevention

- Enable IPS/IDS in order to detect evil acts.
- Scan packets in real time to prevent attacks.

c. Segment Isolation

- Have discrete sensitive networks disaggregated.
- limit access to servers and administrations.

d. Logging and Monitoring

- allow the recording of logs of audit trails and troubleshooting.
- Forwarded logs to a central location monitoring server.

4. Wireless Access Point (WAP) Administration

WAPs provide us with Wi-Fi protection within the campus

a. SSID Setup

- Develop different SSIDs between the staff and guests.
- Isolate internal and visitor traffic.

b. Security Settings

- Authentication with WPA3 or enterprise-grade is available.
- Usage of strong passwords or access by use of certificates.

c. Radio Optimization

- Use as few tuning power and channels as possible to reduce interference.
- strive to achieve uniform coverage everywhere and high throughput.

d. Management Integration

- Enrol WAPs in the central controller or server.
- Implement a standardized template and thus APs seem homogeneous.

5. Server Configuration

a. The backbones are servers that contain authentication, storage, and apps:

Directory and authentication services key directory and authentication services utility directory and authentication services directory and authentication services 6.0 key directory and authentication services utility directory and authentication services probatory directory and authentication services utility directory and authentication services directed probatory directory Enhanced pipe directing probe power affecting the utility of the directory and authentication services key directory and authentication services consumed Do-Not-Resuscitate requests directory react brace directory react brace directory/.LNUTIL wild; passer brace directory/.LNUTIL controlled passer brace directory/.LNUTIL controlled; showing directory confirmation marker directory/.LNUTIL computer

- configure Active Directory or LDAP for user accounts.
- Authorization with switches and routers.

b. Azure file and Application services.

- Sharing folders using role permissions.
- Operate internal applications, web applications, or databases.

c. Network Services

- Deploy DNS, DHCP, and monitoring agents.
- Secure access control of the interfaces of administration.

d. System Hardening

- Have the OS persistently patched and updated.
- Turn off all unnecessary services in order to reduce vulnerabilities.
- It should run host-based firewalls and implement security policies.

6. Configuration of Management and Monitoring.

Gear is maintained in good condition through central monitoring. k Find all the routers, switches, firewalls, and servers, and put them on the monitoring platform.

- Alarm on failed linkage, resource peaks, and security breaches.
- Audit and analysis Store logs centrally.
- Automatic maintenance of device setups.

INTERNET CONNECTIVITY

Therefore, campus networks are all about possessing a robust internet connection. You need a high-speed and quality internet connection to be in a position to chat with the other students, post assignments, and all the other things available online without being left behind the firewall, which makes everything now slower. In simple terms, you should ensure that your connection is fast, safe, and effective in all class assignments, research, and library sources. Among them: ISP connection, placement of the edge router, firewalls, DMZ, routing, backups, VPNs, DNS/DHCP, and monitoring everything [11][12].

1. Connection to Internet Service Provider

It is only half the way to having a good setup unless you have a good ISP to connect you using a dedicated line. At one side of the line is an edge router that takes care of all the external traffic.

1.Key considerations:

- Low latency implies that group project video calls remain clear.
- Adequate bandwidth to maintain campus load (e.g., lecture streaming, giant uploads).
- Uses either BGP, common, or static Internet Service Provider protocols.
- SLA guarantees the availability, low latency, and few packet losses [13].

Example Implementation:
Fibre-optic would be the optimal choice in terms of speed and reliability. Likewise, the edge router obtains public IPs and purposes default paths to the ISP.

2. Role of the Edge Router

The gateway of the campus, which does route and security.

a. Routing The router directs traffic from the VLANs to the ISP.

It also controls web, email, and other types of network traffic.
It allows easy connections through predefined default and static routes.

b. Traffic Translation

- Uses NAT to hide internal IPs.
- increases the availability of IP space to all the online students.
- Uses PAT when there are numerous users to a single IP.

c. Policy Enforcement

- Develops firewalls to restrict the level of traffic that will strike the network.
- Determines accessibility of what, when, and where.
- Checks QoS of mission control operations such as VoIP or lab work [12].

3. Security Perimeter as a Reciprocal Defence Firewall.

To provide additional protection, there is a next-generation firewall between the firewall and the edge router.

a. Access Control Policies

Only permits outgoing paid traffic (web, email, VPN) and stops incoming traffic randomly to forestall attacks.

- Active Offloads DMZ services make classroom servers not at risk.

b. Threat Detection

It has IDS/IPS to identify suspicious activity.

- Blocks malware, phishing, DoS, etc. [13].

c. Logging and Monitoring

- Keeps major event logs.
- Notifies you instantly of the happenings.
- Includes the use of SIEM in the process [14].

4. Adoption of the Demilitarized Zone (DMZ).

The DMZ is a partial barrier of services, which maintains the central net.
Typical DMZ services:

- Public web servers.
- External-facing app servers.

It is teamwork gadgets or email gateways.
Design Considerations:
Firewalls are used to provide stringent filtering between the DMZ and the internal networks.

Only inbound traffic is permitted to DMZ servers, and outbound traffic is controlled. In cases where the DMZ server is compromised, then the core is safe [11][15].

5. Internet Traffic Control

Campus traffic is controlled with the help of smart technology.

a. Bandwidth Control

Context Practices Critical applications take priority (VoIP, ERP, CRM).

- It regulates the use of web-video or social media.
- Has the benefit of reducing the congestion during peak lab hours.

b. Content Filtering

- Blocks Internet suspects or inappropriate websites.
- Increases productivity and reduces security expenses.
- McCloud service interconnects with cloud filtering.

c. Redundancy Options

Provides a second ISP in case the first one is lost or as a load balancer, and deploy two WAN routers so that we can be connected to your study without disconnecting.

- Dynamic routing is an automatic load sharing of traffic [12][13].

6. VPN and Remote Access

In the mobile student and distant study location, Site-to-Site-VPNs are used to link the main campus and the branch sites.

- Client to Site VPNs allow the students to jump securely.
- IPsec or SSL/TLS encryption.

VPN settings are compatible with firewall settings [14].

7. DNS and DHCP Integration

Internal DNS names the names of the apps within the campus. DHCP assigns dynamic IPs. Edge devices are able to support outside queries with fast throughput to accelerate the lookups in order to provide a security layer [15].

8. Serial Checking and Repairing.

Status informs you of inequity connected with health, bandwidth consumption, and delay, and monitors the tracks of anomalies and firewall wellness.

- Notifies of suspicious traffic, network issues, or attacks.
- Automates the compliance reports and debugging logs [11][15].

SERVICES AND APPLICATIONS

Here I shall briefly describe the key network services and applications we have configured to ensure that everything is working well- centralized management, secure access, and efficient resource sharing. These services are the foundation of the network functionality as they support internal business processes as well as the business fronts. Scalability, reliability, and security of the various departments are guaranteed through proper organization and configuration [16][17].

1. Core Network Services

The campus network operates smoothly on core services, which include device comms, IP stuff, and user identities:

a. Domain Name System (DNS)

It resolves hostnames that are readable by people into IP addresses and manages internal applications and the names of servers.

e.g., name recognition within and outside the campus, and convenience to the resources.

Additional DNS servers have it running 24/7 [16].

b. Dynamic Host Configuration Protocol (DHCP)

- Issues IPs and network parameters automatically.
- Provides homogeneous configuration on gateways, subnets, and DNS servers.
- prevents IP conflicts and saves administrative time.
- DHCP failover ensures that service is maintained in case one of the servers fails (crashes) [17].
- c. Authentication and Directory Services.

The central authentication is the one where only legitimate people can enter.

2. Web and application services.

We are efficient, internally and in contact with the outside world, with the help of web and app services

a. Internal Web Services

- It is used for hosting department applications.
- Tools of collab and project management are centrally available.
- Sensitive data is filtered by access policies to enable the appropriate individuals to access it.

b. Public Web Services

Websites and portals are taken care of in a safe manner.

- Based on the DMZ to keep internal gear under the carpet.

Load balancing and redundancy provide excellent uptime and quick responsiveness [18].

c. Application Services

Internal ERP, CRM, or custom applications exist as a dedicated server.

- ACLs and firewalls regulate access.
- Storage and QoS are high-performance to ensure snappy services [19].

3. File and Data Services

a. File Sharing

Central file Servers have docs, dept resources, and backups.

In permissions, the data is only visible to a few users with authorizations.

- Eliminates the presence of duplicate data.

b. Storage Management

Rational storage configurations align with the project and project requirements.

RAID prevents data loss and provides auto-backups.

- Storage health and capacity alert tools are monitored [16].

c. Database Services

Central databases are implemented in business-critical apps, Clustering, and replication. Clustering is highly available and provides disaster recovery.

4. Communication Services and Collaboration

a. Voice over IP (VoIP)

- Favours campus telephones, both indoors and out.
- Call traffic is priority-based with a dedicated voice VLAN.
- Combines voicemail, conferencing, and directory services.

b. Email and Messaging

- Campus and external mail central.
- Instant messaging, file sharing, and scheduling are available in team platforms.
- Constant availability of email servers is redundant.

c. Video Conferencing

- Increases video conferencing and customer calls.
- Video is smooth with bandwidth control and QoS.

- Use calendars and collaborative applications to work on sessions that are easy to use [18].

5. Wireless Network Services

a. Corporate Wireless

- Wi -Fi on campus is secure and accessible to everybody working on campus.
- Enterprise/802.1X WPA3 provides security of sensitive data.
- All is taken care of by the central management [19].

b. Guest Wireless

Guests are assigned a different network slice.

- Detached internal resources to secure.
- Provides easy internet services without any interference with the security of the campus.

6. Security and Management Services.

a. Monitoring and Logging

Network monitoring monitors the health of devices, trends in traffic, and services on demand. Switch, firewall, router, and server logs are gathered to be used in troubleshooting and analysis. Weird actions or failures are alerted to admins [16][20].

b. Admission Control and Authorization.

- Allowances for who and what devices have network access due to roles, VLANs, and authentication.
- By policies, the levels of required security are enforced within the campus.
- Embedded firewall, VPN, and Wi-Fi policy: End-to-end safe.

c. Backup and Recovery

Auto back save application data, configuration files, and user data, and disaster recovery plans recover the necessary services within a short time.

Backup retention policies are timed and aid in compliance and continuity.

d. Load Balancing/Redundancy.

Web, email, and app services are configured for high availability. The load balancers spread out the traffic in the same manner, eliminating overloads and downtime. Hardware redundancy removes the risk of service failure in the event of component malfunction [17][20].

7. Testing of Services

- Functional and performance testing: ensures that all works well
- Name resolution and check IP assignment.
- Check file server, web applications, and internal portal access.
- Diagnose wireless coverage and authentication.
- Video conference, email, and validate VoIP.
- Stress test load balancers and failover

The successful tests allow verifying that services work in accordance with operational needs and provide stable performance throughout the network [16][18].

8. Cloud and Remote Services

The cloud-based email, collab, or file storage can be used to support campus services. Secure VPN or Direct connection accesses cloud apps in a secure manner. The Hybrid AAA systems achieve scalability without involving any infrastructure in the campus [17][19][20].

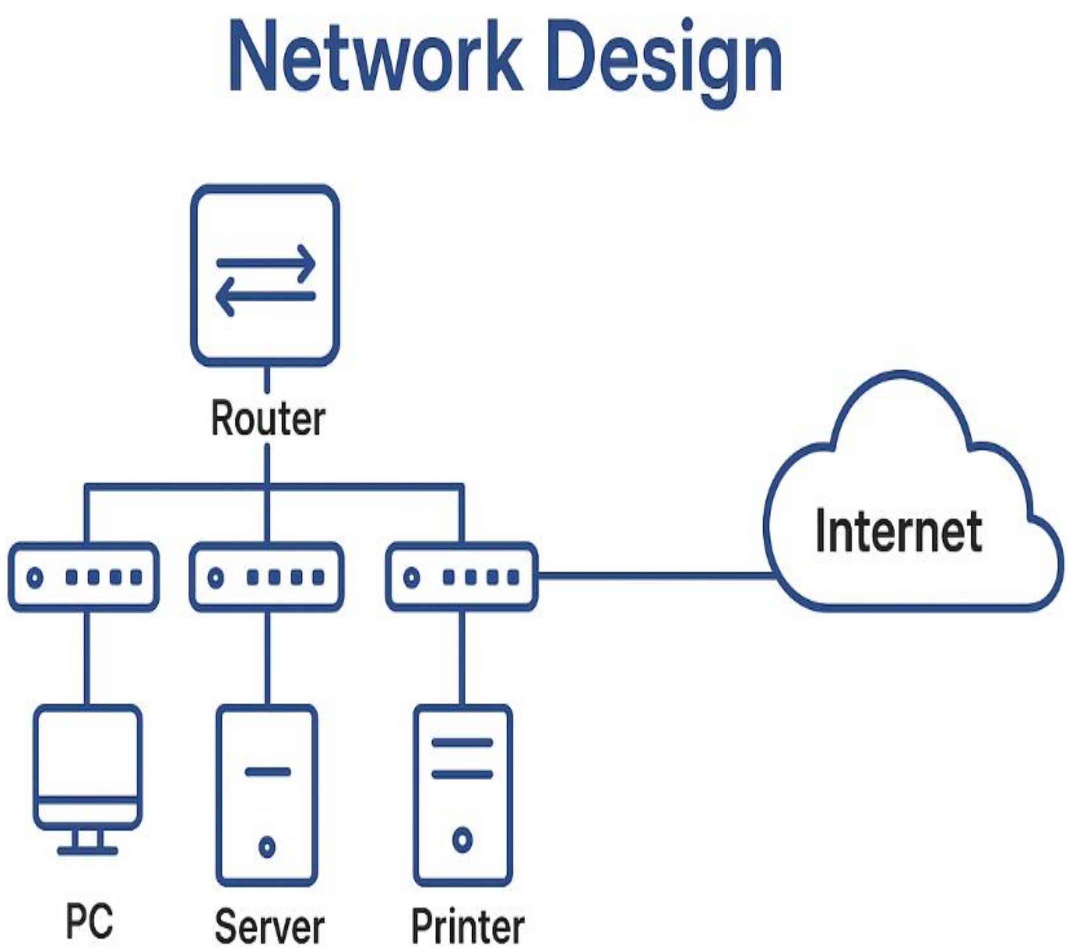
5. TESTING, VALIDATION, AND TROUBLESHOOTING.

5.0 Introduction

Once we put up and set up a network, we need to ensure that it is functioning well, secure, and that it does not drop the ball. This is why we do a lot of testing, validation, and troubleshooting- this keeps all the bits and pieces operating, the services in operation, and the performance on track.

- Network components really work - no dead ends.
- Services are reliable and non-quit.
- Security remains very rigid regarding any potential attacks.

Problem testing in everyday use. In day-to-day operations, validation will prove that we are actually achieving the design objectives, and troubleshooting will provide us with a clear schedule on how to rectify any problems that occur. They are collectively able to safeguard reliability, scalability, and resilience [21][22].



5.1 Testing

Testing involves the process of ensuring that all the functionality of the network is as it is supposed to be during rollout and after.

5.1.1 Connectivity Testing

We must ensure that communication between devices takes place throughout the campus network.

- The different departments or the VLAN within the same one.
- Interdepartmental or inter-VLAN between the external Internet resources.

Practical Scenario: When an employee is tying up the computer, the laptop must connect smoothly to the company file system, email, and external sites. The hint of failure may indicate some misconfigurations, hardware errors, or defective cables [21].

5.1.2 Testing of the Devices and Interfaces.

It ensures routers, switches, APs, and Servers are online.

- The machines are turned on and active.
- Interfaces: Networks are connected and operational.
- The devices can communicate with the parts of the network.

Practical Use: When the switch linking several PCs in the office is not carrying the traffic, we isolate the switch and the cabling to determine the fault [22].

5.1.3 Performance Testing

It tests network maintainability.

Bandwidth and throughput: what data receives passage.

Cross delays and round-trip time: quantify delays, particularly VoIP/video delays.

Practical Example: Browsing, transferring files, and video calls (when there are peak hours) should be nearly instant. Bottlenecks indicate that it requires additional bandwidth or hardware adjustments [23].

5.1.4 Security Testing

We maintain the system closed by verification of the perimeters.

- restriction of access and firewalls.
- VPN access for remote users.
- Scans of devices and services (vulnerability).

Practical: Simulated bad-actor attempts ensure that they are not allowed inside the firewall and that they do not access our data [24].

5.2 Validation

Validation enables checking whether we are really working on the design and fulfilling business requirements.

The following section addresses the authentication of IP addresses and subnets by confirming they are valid and unvarying over time. 5.2.1 Validation of the IP address and subnet. It ensures all the devices receive appropriate IP addresses and they are not in collision.

Practical Case Study: The devices of different departments can only communicate via approved paths, which enables traffic to maintain order and be safe [21].

5.2.2 Routing and Path Validation

Data has been confirmed to take the right paths.

- Smooth supply, no losses or undesired time.
- Redundancy provides possible alternatives in the event of a failure

Pragmatic Case Study: A good route can be used to maintain the requests to the backup server in a different building when the first server is overwhelmed [22].

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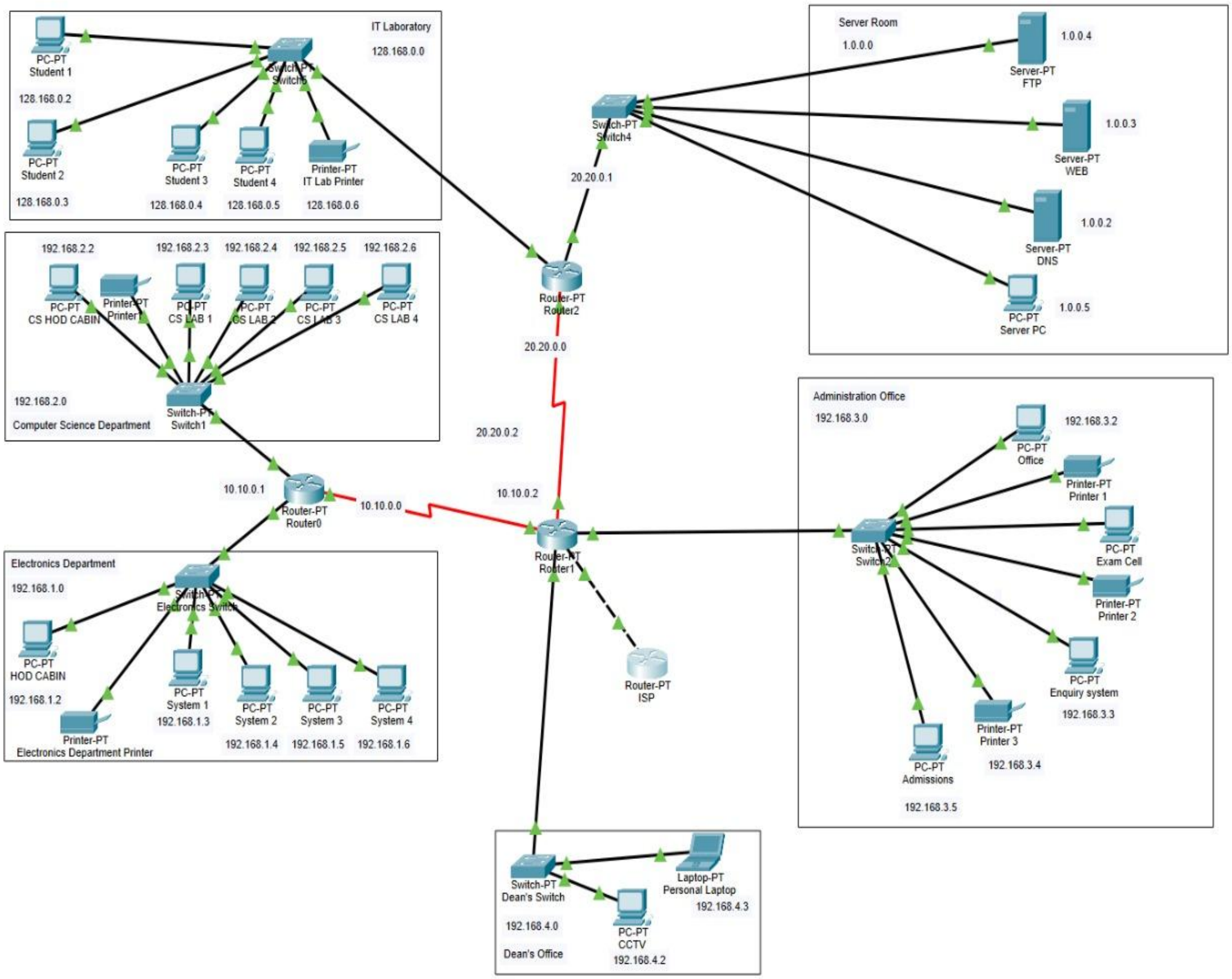


Pragmatic Case Study: A good route can be used to maintain the requests to the backup server in a different building when the first server is overwhelmed [22].

5.2.3 service and application validation

Make sure that core services remain solid.

- VoIP/video, file sharing, internal web apps, email.

Practical Case: Workers will have access to shared documents, make VoIP calls, and receive emails without any hassle [23].

- Validity at Redundancy and Failover.
- Double-check backup systems.

- Processes Backup links on critical paths.
- Back-up servers of important services.
- Problem-free automatic fall-out in case of outages.

Practical Case: On failure of the major Internet, traffic flows through the secondary link without missing a step [24].

5.2.5 Compliance and Standards Checking.

Ensure that we are following all the regulations.

- Proper cabling techniques.
- Account control and defence in depth protection.
- Regulatory compliance.

Practical Case: The practical mapping of departments can address cybersecurity policies and prevent data leakage and legal complications [25].

5.3 Troubleshooting

A strict set of guidelines that will be used to find, diagnose, and resolve network issues.

5.3.1 Problem Identification

- Be aware of slow response, inability to connect, or intermittent connection.
- Gather user reports.
- Monitoring anomalies in scanners.

5.3.2 Isolation and Diagnosis

- Isolate the affected parts of the network by subdividing the network.
- Find hardware, software problems, or configuration.
- focus on the most vital issues.

Practical Case: In case one of the departments fails to connect with the file server, the administrator isolates switches, routers, and connections until the offender appears [21].

5.3.3 Tools and Techniques

- Automation of traffic patterns and bandwidth.
- Delays at the spot, packet drop-outs, or interruption of services.
- Make a comparison with how things should be.

Practical Case: Peak-hour slackness is then tracked down to congested devices or connections and upgraded [22].
This study is carried out in terms of what has already been written by people. We go through all the papers, countercheck what each one believes, and then decide how the problem can be solved most favourably.

5.3.4 Resolution and Verification.

Therefore, in this study, the intention is to sift through the literature and discuss our findings in order to select the most appropriate solution that may be in line with our observations in the modern world. Fixes assist you in working around some hardware glitches temporarily, adjusting your settings, or restoring a service to its former state.

- Test that the service is implemented to the full extent.

Real Life example: installing a faulty switch with a new one restores the connection to all devices [23]. Documentation and report. This section conveys that management is expected to come across as transparent, truthful, and inclusive.

5.3.5 Documentation and Reporting.

This part expresses that the management should sound transparent, true, and inclusive.

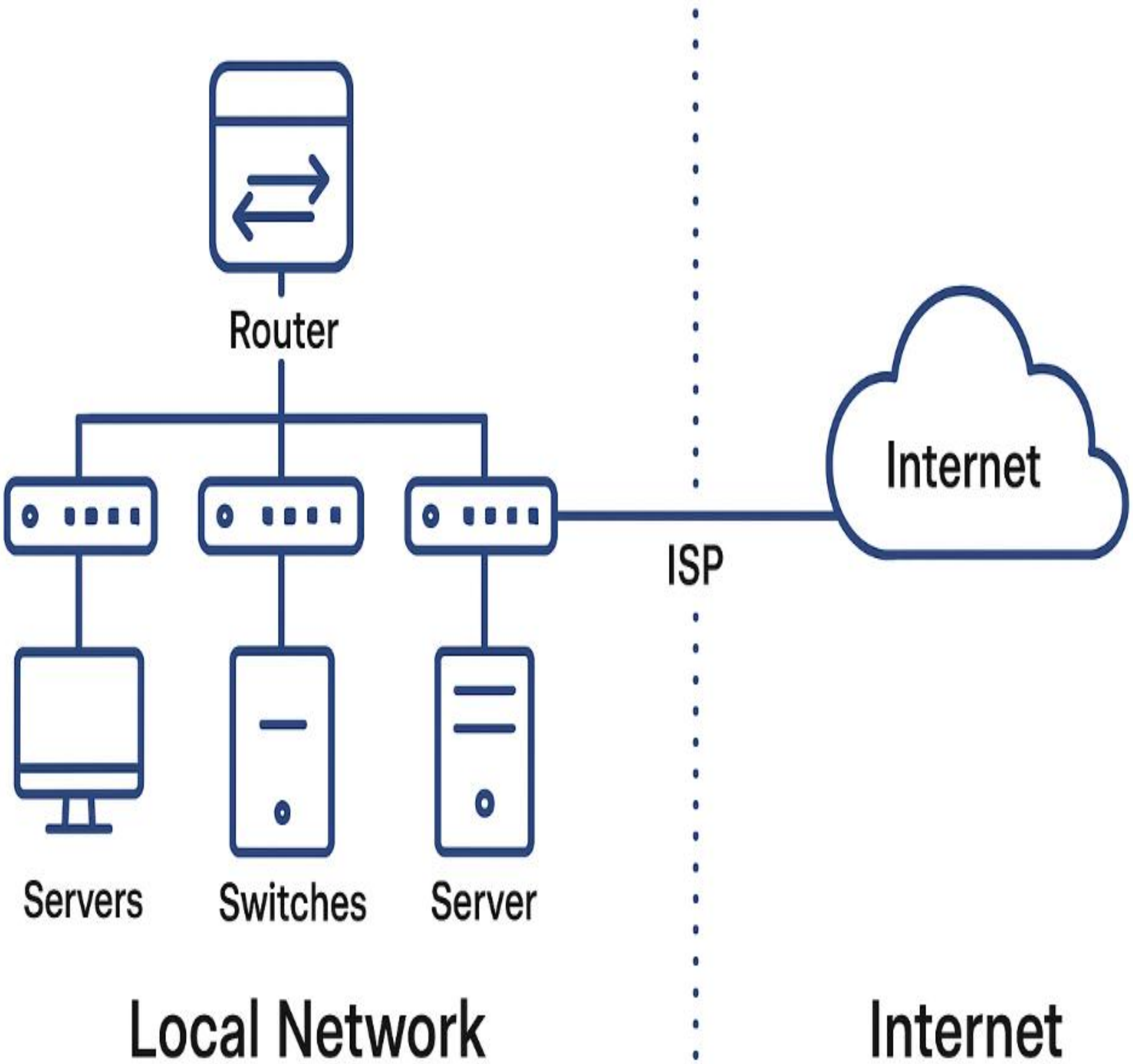
- Make records to help do fast repairs in the future.
- Utilize them to implement prevention.
- Present a complaint history on compliance that is audit-ready.

Real-life Hyphenises: Symptoms, root cause, and fix, as well as lessons learned, are recorded in logs [24][25].

5.4 Case Studies

1. Configuration of VLAN: VLAN configuration error terminated one-way access to shared drives. Restoration of connectivity was achieved by repairing the config.
2. Bandwidth Bottleneck: File downloads were slowed down during peak time. The solution was to add bandwidth and provide a priority to the traffic.

3. Security Breach Simulation: Firewall prevented inappropriate operations, and this proved our security settings.



5.5 Best Practices

- Post-deployment performative tests.
- Test all the changes before implementing them.
- Have backup and recovery on standby.
- Has to keep track of performance.

- Document setting, topologies, and troubleshooting measures.
- Devices, VLANs, and IP addresses should have similar names [21][25].

Summary

In essence, all this network design affair is simply concerned with the establishment of a system that would cater to all the current demands of the organization with the aim of ensuring that everything goes safely, levels, and of the highest standards. The following are the main points that attracted my attention.

- To enhance its performance and efficiency, the company incurs a ton on sports sponsorships on top of advertisements and promoting sports (Shrunk, 2006).
- High performance and efficiency: The firm spends a significant amount of money on sport sponsorship, advertisement, and popularisation of the sport activities, which makes it a high performer (Shrunk, 2006).

We are talking about a network brought down to the proper layers, the addition of VLANs, the control of bandwidth like an expert, and the application of QoS rules to ensure the network flows smoothly and every department operates without any hiccup.

Robust Security:

It has all firewalls, access-control processes, wireless seclusion, safe VPN, and DMZ to lock the internal resources and only accept the appropriate persons from the external.

Reliability and Resilience:

We have introduced a bit of unnecessary added material, installed an auto-switch, implemented alternative power, and kept an eye on it at all times, such that the service is maintained with hardly any shutdowns.
The cloud-based solution is easy to scale up in case the user count is increased.

Scalability and Flexibility:

They have designed IP addresses, centralized control, and a modular architecture that enables them to add capacity and change along with organizational requirements, without rewriting the architecture.

Manageability:

The vendor maintains a large and well-organised collection of documented and recommended practices.

Operation Support:

The vendor provides well-documented and broadly organized best practices.
The kit takes care of the administration within a single location and real-time application, backup, and recovery when a breakage occurs, and generally keeps the admins busy and the network glitch-free.

Validated Functionality:

Test, test, test, and adjust until all devices, services, and applications are up and functioning, and are satisfactory in performance and security.
Business Operations: The Business Operations Support System currently being used should not allow repetitive business processes that could compromise the operational efficiency of the organization within the market.

The Business Operations:

The support system employed in the business activities must ensure that operations in the marketplace are not repeated to create threats to the efficiency of the operations.
The network will be configured to address the daily activities, team-to-team processes, and both cloud integration and external connectivity without jeopardizing information security and accessibility to the user.

To be brief, we have created a secure, scalable, robust, and fully controllable environment through good planning, proper configuration, constant testing, and constant monitoring that can conveniently accommodate current and possible future operations.

References

[1] Radosavovic, I., Kosaraju, R. P., Girshick, R., He, K., & Dollár, P. (2020). Designing network design spaces. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 10428–10436.
[2] Johnson, D. S., Lenstra, J. K., & Kan, A. H. G. R. (1978). The complexity of the network design problem. Networks, 8(4), 279–285.
[3] Farahani, R. Z., Miandoabchi, E., Szeto, W. Y., & Rashidi, H. (2013). A review of urban transportation network design problems. European Journal of Operational Research, 229(2), 281–302.
[4] Ceder, A., & Wilson, N. H. M. (1986). Bus network design. Transportation Research Part B: Methodological, 20(4), 331–344.
[5] Park, P., Ergen, S. C., Fischione, C., Lu, C., & Johansson, K. H. (2017). Wireless network design for control systems: A survey. IEEE Communications Surveys & Tutorials, 20(2), 978–1013.
[6] Kim, B. C., Hong, J.-Y., Wallace, G. G., & Park, H. S. (2015). Recent progress in flexible electrochemical capacitors: electrode materials, device configuration, and functions. Advanced Energy Materials, 5(22), 1500959.
[7] Schüpbach, A., Baumann, A., Roscoe, T., & Peter, S. (2012). A declarative language approach to device configuration. ACM Transactions on Computer Systems, 30(1), 1–35.
[8] Kuutti, A., Dvoryanchikova, A., Lobov, A., Martinez Lastra, J. L., & Vantera, T. (2012). A device configuration management tool for context-aware systems. IEEE 10th International Conference on Industrial Informatics, 10–15.
[9] Liang, J., & Wang, D.-W. (2022). Design rationale and device configuration of lithium-ion capacitors. Advanced Energy Materials, 12(25), 2200920.
[10] Zhang, C., Chen, P., & Hu, W. (2016). Organic light-emitting transistors: materials, device configurations, and operations. Small, 12(10), 1252–1294.
[11] Haythornthwaite, C. (2005). Social networks and Internet connectivity effects. Information, Community & Society, 8(2), 125–147.
[12] Strover, S. (2001). Rural internet connectivity. Telecommunications Policy, 25(5), 331–347.
[13] Salac, R. A., & Kim, Y. S. (2016). A study on the internet connectivity in the Philippines. Asia Pacific Journal of Business Review, 1(1), 67–88.
[14] Hjort, J., & Tian, L. (2021). The economic impact of internet connectivity in developing countries. Annual Review of Economics, 17.
[15] Bahinting, M. A. P., et al. (2022). Stronger than the internet connectivity: a phenomenology. Psychology and Education: A Multidisciplinary Journal, 2(6), 1–19.
[16] Birman, K. P. (2005). Reliable distributed systems: technologies, web services, and applications. Springer.
[17] Youssef, A. E. (2012). Exploring cloud computing services and applications. Journal of Emerging Trends in Computing and Information Sciences, 3(6), 838–847.
[18] Leymann, F. (2003). Web services: Distributed applications without limits-an outline. BTW 2003--Datenbanksysteme für Business, Technologie und Web, Tagungsband der 10. BTW Konferenz, 2–23.
[19] Balakrishna, C. (2012). Enabling technologies for smart city services and applications. 2012 Sixth International Conference on Next Generation Mobile Applications, Services and Technologies, 223–227.
[20] Lei, H., Sow, D. M., Davis, J. S., Banavar, G., & Ebling, M. R. (2002). The design and applications of a context service. ACM SIGMOBILE Mobile Computing and Communications Review, 6(4), 45–55.
[21] Leclercq, C., Mabo, P., & Daubert, J. C. (2008). Troubleshooting. Cardiac Resynchronization Therapy, 324–355. Wiley Online Library.
[22] Wingate, G. (2016). Practical Troubleshooting. In Pharmaceutical Computer Systems Validation (pp. 388–393). CRC Press.
[23] Dholakiya, S. L., & Barile, F. A. (2013). Alternative methods for ocular toxicology testing: validation, applications, and troubleshooting. Expert Opinion on Drug Metabolism & Toxicology, 9(6), 699–712.
[24] Parrella, J., Raghunath, B., & Lutz, H. (2015). Troubleshooting. In Ultrafiltration for Bioprocessing (pp. 181–201). Elsevier.

[25] Zhao, Z., Chen, M., Matheson, D., & Stone, M. (2016). Online experimentation, diagnosis, and troubleshooting beyond AA validation. 2016 IEEE International Conference on Data Science and Advanced Analytics (DSAA), 498–507. IEEE.

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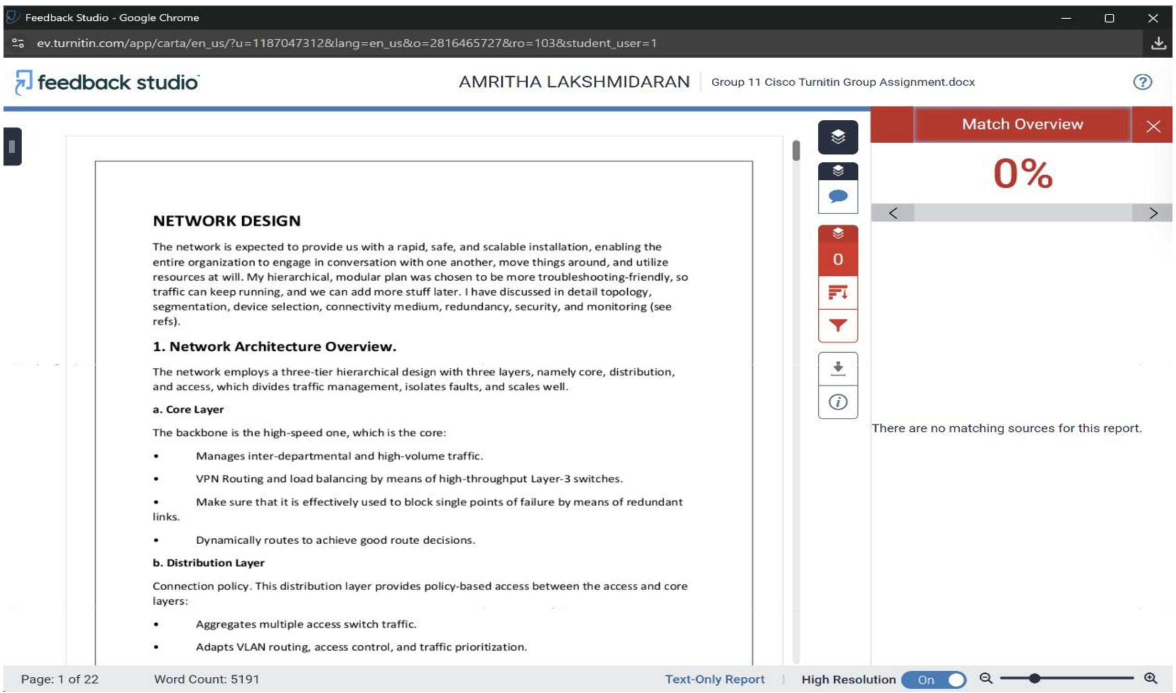

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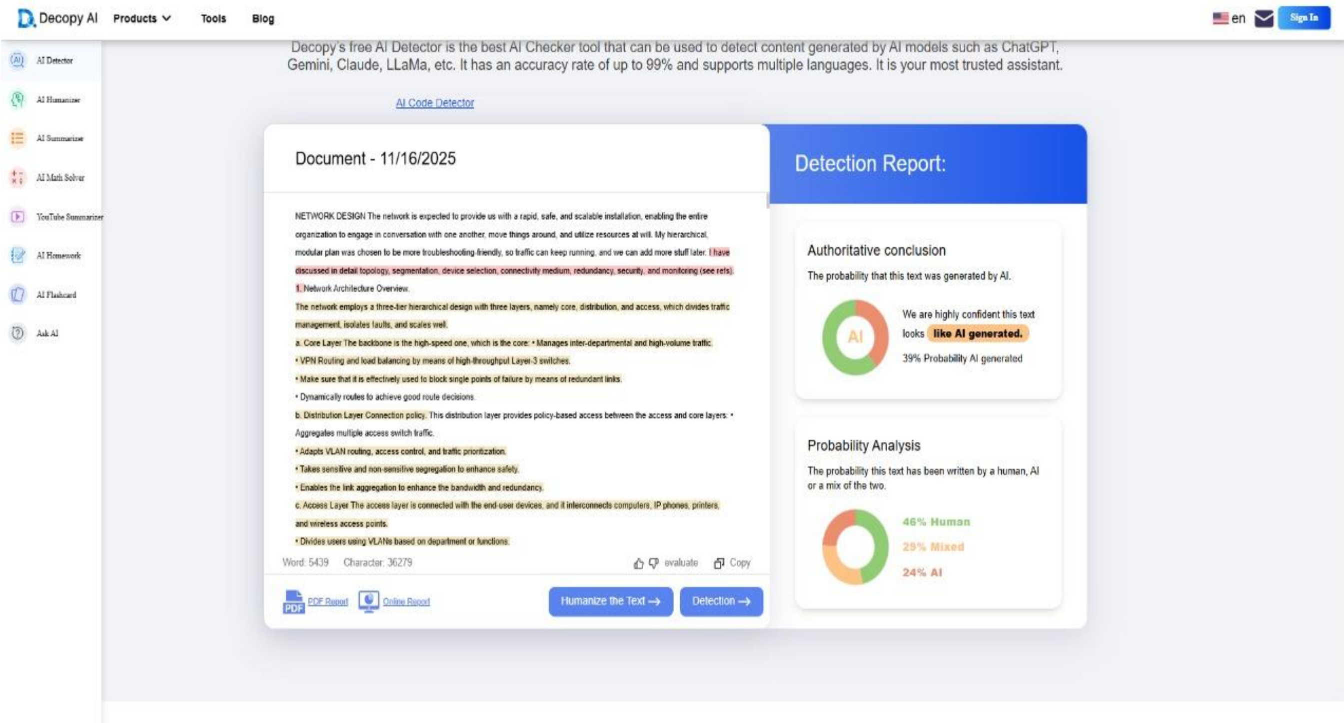
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Detection Report:
Authoritative conclusion
The probability that this text was generated by AI.
We are highly confident this text looks like AI generated.
39% Probability AI generated
Probability Analysis
The probability this text has been written by a human, AI or a mix of the two.
46% Human
29% Mixed
24% AI
Humanize the Text
Detection

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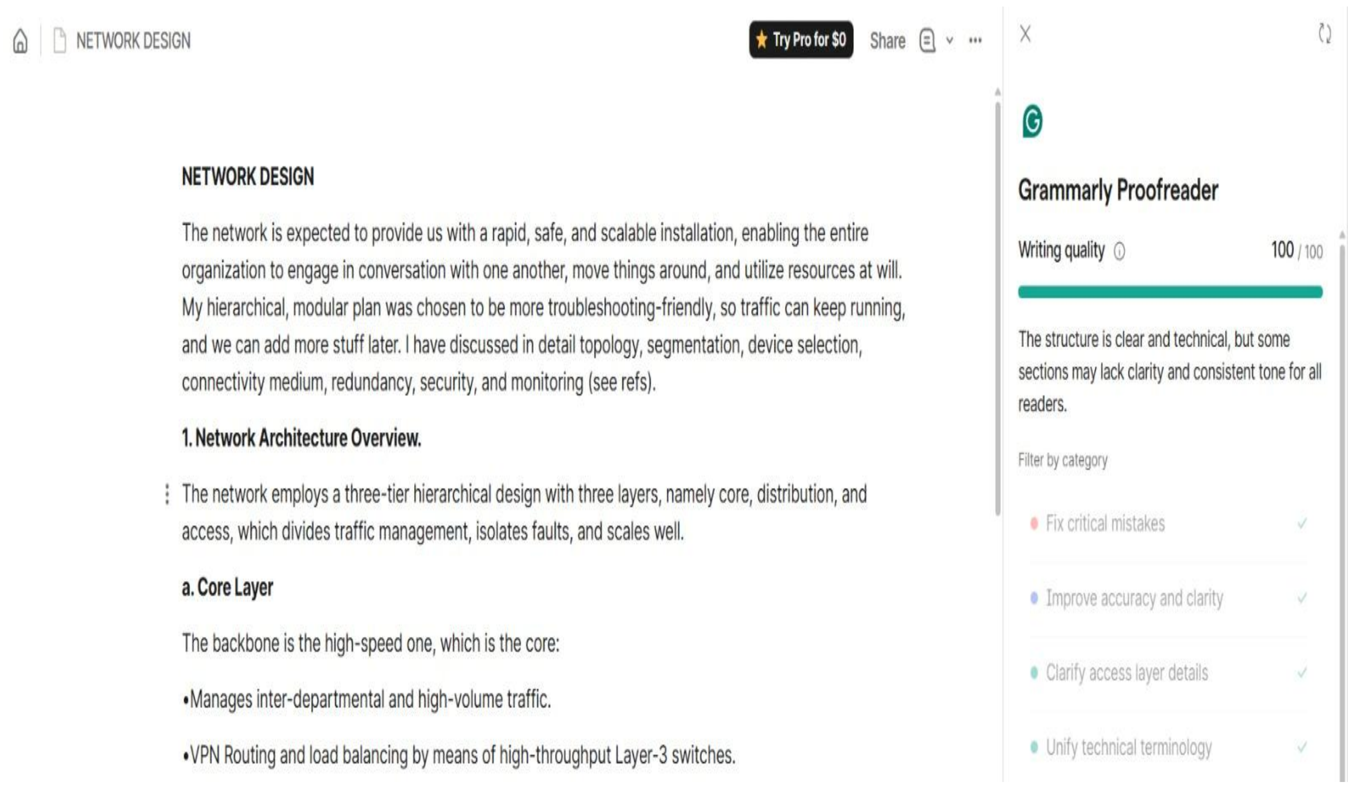
NETWORK DESIGN
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NETWORK DESIGN
The network is expected to provide us with a rapid, safe, and scalable installation, enabling the entire organization to engage in conversation with one another, move things around, and utilize resources at will. My hierarchical, modular plan was chosen to be more troubleshooting-friendly, so traffic can keep running, and we can add more stuff later. I have discussed in detail topology, segmentation, device selection, connectivity medium, redundancy, security, and monitoring (see refs).
1. Network Architecture Overview.
The network employs a three-tier hierarchical design with three layers, namely core, distribution, and access, which divides traffic management, isolates faults, and scales well.
a. Core Layer
The backbone is the high-speed one, which is the core:
•Manages inter-departmental and high-volume traffic.
•VPN Routing and load balancing by means of high-throughput Layer-3 switches.
Grammarly Proofreader
Writing quality
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The structure is clear and technical, but some sections may lack clarity and consistent tone for all readers.
Filter by category
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